AMERICAN INTERNATIONAL UNIVERSITY – BANGLADESH
Faculty of Engineering

Course Name: Introduction to Electrical Circuits

Semester: Spring 2025-26 **Term:** Mid **Quiz:** 02 **Total Marks:** 20 **Time:** 40 Minutes

Question Mapping with Course Outcomes:

Item	COs	POIs	K	P	A	Marks	Obtained Marks
Q 1-4	CO2	P.a.2.C3	K2			4 x 5	
					Total:	20	

Student Information:

Student Name:	**Student ID:**
Section: K	**Department:**

1. Using Thevenin's Theorem in Figure 1, calculate the current passing through the resistor R_4. **[5]**

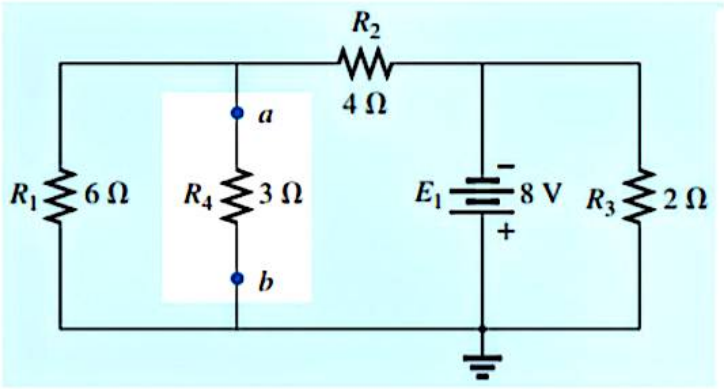


Figure. 1: *Circuit diagram for question 1*

2. Find the Norton equivalent circuit for the network in the enclosed area in Figure 2. Also, find the value of R_L for maximum power to R_L and determine the maximum power to R_L. **[5]**

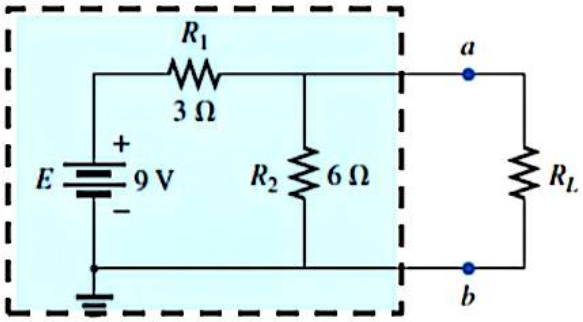


Figure. 2: *Circuit diagram for question 2*

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3. Using superposition theorem, find the voltage V_3 for the following network in Figure 3. **[5]**

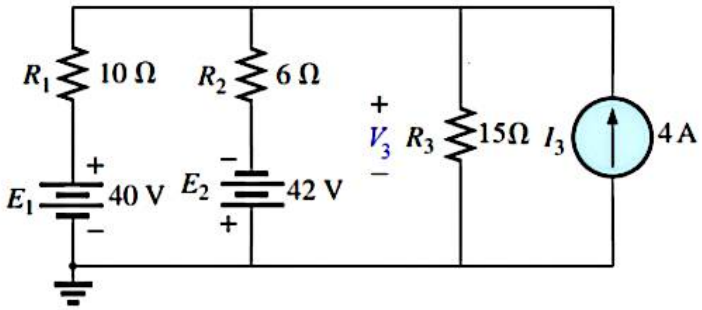


Figure. 3: *Circuit diagram for question 3*

4. Find the current through each resistor for the following networks in Figure 4.
Use Mesh analysis. **[5]**

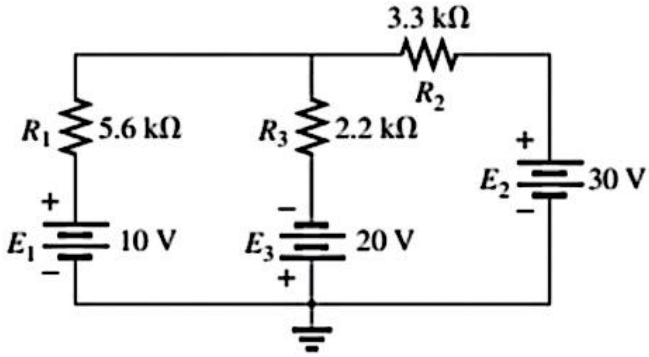


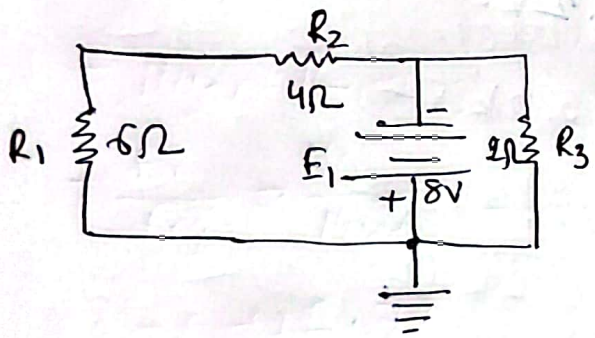
Figure. 4: *Circuit diagram for question 4*

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Ans to the Q NO-01



using voltage divider rule

$$V_{R1} = 8 \times \frac{R_1}{R_1+R_2}$$

$$= 8 \times \frac{6}{6+4}$$

$$= 4.8\ V$$

$$R_s = R_2+R_3 = 4+2 = 6\ \Omega$$

$$R_{th} = \frac{1}{R_1} + \frac{1}{R_s}$$

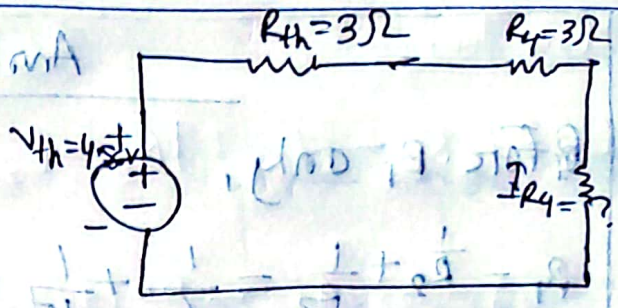
$$= \frac{1}{6} + \frac{1}{6}$$

$$= 3\Omega$$

$$I_{R_4} = \frac{V_{th}}{R_{th} + R_4}$$

$$= \frac{4.8}{3+3}$$

$$= 0.8\,A$$



Ans to the Q NO-02

$$R_{N_2} = \frac{R_1 \times R_2}{R_1 + R_2}$$

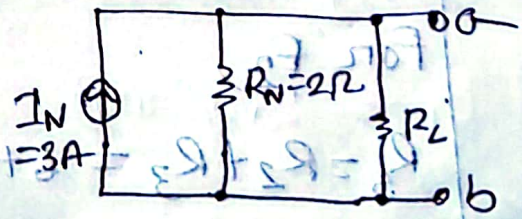
$$= \frac{3 \cdot 6}{3+6} = \frac{18}{9} = 2$$

$$I_N = \frac{9}{3+0} = 3\,A$$

Wekness,

Maximum powere transfere when $R_L = R_N = 2\Omega$

$$P_{max} = \frac{I_N^2 \cdot R_N}{4} = \frac{3^2 \cdot 2}{4} = 4.5\,W$$

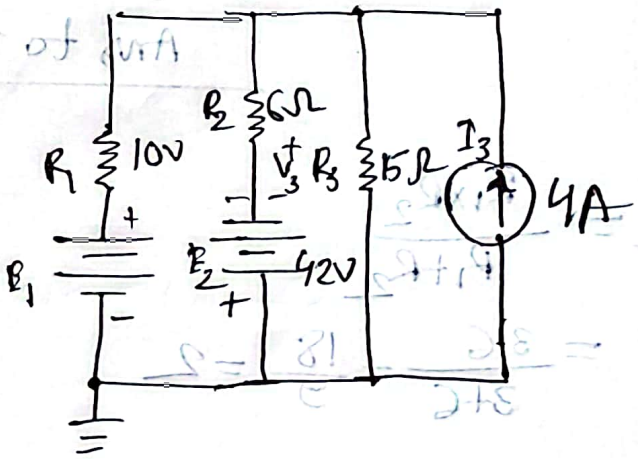


Ans to the QNo-3

For E_1 only,

$$R_T = \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{6} + \frac{1}{15} = 4.286\,\Omega$$

$$R_{eq} = R_1 + R_T$$
$$= 10 + 4.286$$
$$= 14.286\,\Omega$$



$$I = \frac{E_1}{R_{eq}} = \frac{40}{14.286}$$
$$= 2.79$$
$$= 2.8\ A$$

$$I_{R_3} = I_{E_1} \times \frac{R_2}{R_2 + R_3}$$

$$= 2.8 \times \frac{6}{6+15} = 0.8\ A$$

$$V_{R_3} = I_{R_3} \times R_3 = 0.8 \times 15 = 12\ V$$

For E_2,

$$R_S = R_2 + R_3 = 6 + 15 = 21$$

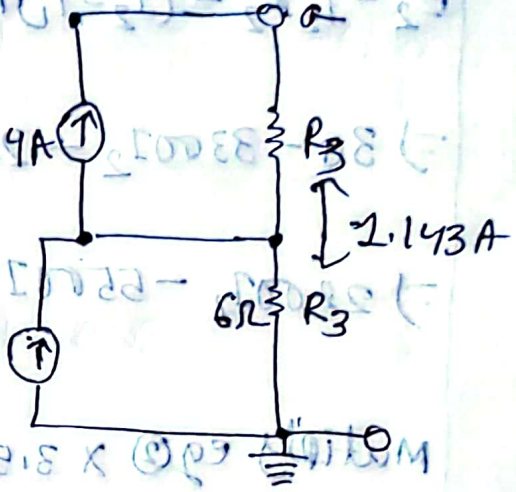
$$I_{E_2} = \frac{E_2}{R_2 + R_3} = \frac{42}{21} = 2A$$

~~$I_{R_3} = 1$~~

$$I_{R_3} = I_3 \times \frac{R_2}{R_2 + R_3}$$

$$= 4 \times \frac{6}{6+15}$$

$$= 1.143 \text{ A}$$



$$V_{R_3} = I_{R_3} \times R_3 = 1.143 \times 15$$

$$= 17.145 \text{ V}$$

$$V_3 = V_{3E_1} + V_{3E_2} + V_{3I_3}$$

$$= 12 + 30 + 17.15$$

$$= 59.15 \text{ V}$$

Ans to the NO – 4

for loop 1,

using KVL

$$E_1 - I_1 R_1 - (I_1 - I_2) R_3 = 0$$

$$\Rightarrow 10 - 5600 I_1 - 2200 (I_1 - I_2) = 0$$

$$\Rightarrow 7800 I_1 - 2200 I_2 = 10 \quad \text{---(1)}$$

for loop 2,

$$E_2 - I_2R_2 - (I_2 - I_1)R_3 - E_3 = 0$$

$$\Rightarrow 30 - 3300I_2 - 2200(I_2 - I_1) - 20 = 0$$

$$\Rightarrow 22.00I_1 - 5500I_2 = -10 \quad \text{---(2)}$$

Multiply eq (2) × 3.545

$$7800I_1 - 19497.5I_2 = -35.45 \quad \text{---(3)}$$

eq (3) − eq (1)

$$-19497.5I_2 + 2200I_2 = -35.45 - 10$$

$$I_2 = 0.00263\ A = 2.63\ mA$$

In eq (1) ⇒

$$7800I_1 - 2200(0.00263) = 10$$

$$I_1 = 2.02\ mA$$

$$I_{R_3} = I_1 - I_2$$

$$= 2.02 - 2.63 = -0.61\ mA \text{ (Aniti clock wise)}$$